

Alexander Huynh

ahuynh23@uw.edu ❖ Bellevue, WA

EDUCATION

Bachelor of Science in Informatics (Data Science Track)

University of Washington | 3.75 GPA

June 2026

Seattle, WA

WORK EXPERIENCE

Database Systems Teaching Assistant

INFO 330: Databases and Data Modeling – UW Information School

Jan. 2025 – Mar. 2025

Seattle, WA

- Led weekly SQL labs and coached 20+ undergraduates on schema design, normalization, and multi-table querying.
- Mentored teams on ER modeling and SQL Server implementation across 10+ projects, reinforcing best practices in referential integrity and indexing.
- Diagnosed query inefficiencies and schema errors, helping students build performant databases aligned with real-world data architecture principles.

PROJECTS

Fraud Detection Pipeline & Risk Scoring (Python, scikit-learn, XGBoost, matplotlib)

Apr. 2024 – May. 2025

- Simulated 10,000+ user sessions and engineered behavioral features (e.g., VPN usage, cart-no-login, short session) to flag fraud risk patterns.
- Trained and evaluated classification models (Logistic Regression, Random Forest, XGBoost) on imbalanced data, achieving ROC AUC > 0.98 and high precision-recall performance.
- Delivered a real-time fraud scoring demo with threshold-based alerts and feature importance plots to support interpretability and operational decision-making.

Recommender System Modeling & Evaluation (Python, NumPy)

Nov. 2024 – May. 2025

- Collaborated with a team of 4 to design a personalized investment recommender using risk tolerance, return satisfaction, product preferences, and holding patterns to match users with high-performing peer portfolios.
- Simulated 1,000+ investor records and benchmarked cosine similarity, Euclidean distance, and weighted scoring to validate peer-matching accuracy.
- Filtered top 10% benchmark-outperforming portfolios and surfaced similar investor strategies to support transparent, satisfaction-driven recommendations.

SQL Data Modeling & Query Optimization (SQL Server)

Apr. 2024 – Jun. 2024

- Designed a normalized relational database for 1,000+ shelter and resident records; implemented ER modeling, indexing, and referential constraints to ensure integrity and performance.
- Authored stored procedures, multi-table joins, and aggregation views to support reporting on capacity, turnover, and resource allocation.

Digital Asset Clustering & Prototype Pipeline (Python, NumPy, PIL, scikit-learn, Power BI)

Feb. 2025 – Apr. 2025

- Clustered 300+ unlabeled 8×8 game assets using PCA and KMeans on engineered visual features (color, symmetry, edge structure).
- Automated duplicate detection and group synthesis to support scalable content filtering, variant generation, and review.
- Delivered Power BI dashboards and CSV summaries to enable batch asset evaluation, visual QA, and rapid iteration cycles.

CO₂ Emission Forecasting with Linear Regression (R, dplyr, tidyr)

Jan. 2024 – Mar. 2024

- Collaborated with 3 peers to integrate and clean 10+ public datasets (energy use, urbanization, fertility, air quality) to predict CO₂ output across countries.
- Trained a linear regression model ($R^2 = 0.95$), selected key predictors, and validated performance with residual analysis and cross-validation.

SKILLS

- **Languages/Tools:** Python, R, SQL, Java, JavaScript, HTML/CSS, Git, Tableau, Power BI
- **Libraries/Tech:** pandas, NumPy, scikit-learn, matplotlib, PIL, SQL Server, Azure Data Studio
- **Core Skills:** Data Cleaning, EDA, Feature Engineering, Predictive Modeling, Clustering, Recommender Systems, Agile Workflow